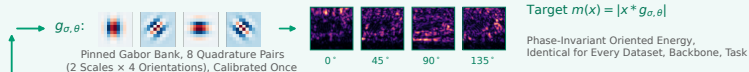


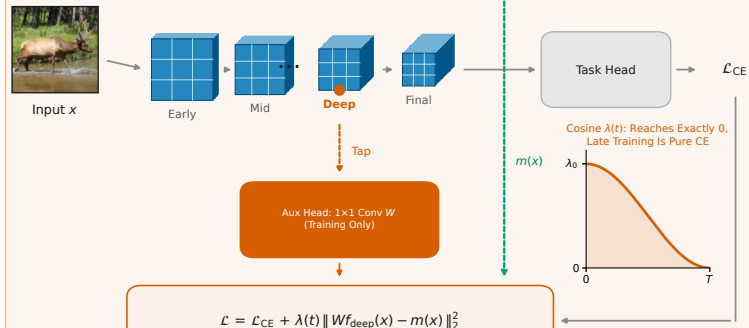
(a) MomentAux in Three Phases: Fixed Source, Fused Training, Plain Deployment

Phase 1 · The Knowledge Source (Fixed and Fingerprinted Before Any Training)



Phase 2 · Fused Training: One Input, Two Losses

Backbone: Four Stages, Unchanged (Spatial ↓, Channels ↑)



Phase 3 · Deployment: W and the Bank Are Dropped



- Classification
- Segmentation
- Detection

One Task Head, Trained for the Task at Hand: the Same Bank, Tap and Schedule Serve All Three; FLOPs and Parameters Identical to the Baseline

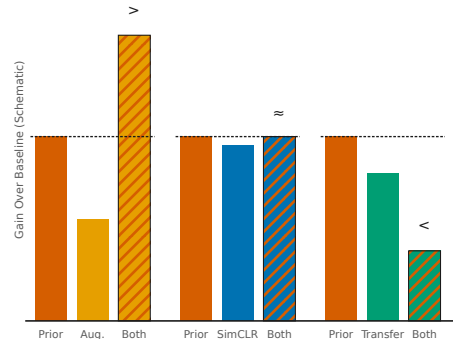
(b) Fusion Outcomes: What Two Sources Can Do

Bars Schematic; Dashed Line: Better Single Source

Stack

Substitute

Interfere



Prior + DeiT Aug.:
+21.0 vs +13.3 Alone
(ViT, C100@10%)

Prior + SimCLR
= SimCLR Alone

Prior on ImageNet Init:
-16 at $\lambda_0 = 1.0$,
= 0 at $\lambda_0 \leq 0.3$

$$\Delta = G + \text{readout}$$

G : Frozen-Feature Gain (Linear Evaluation); readout: What the Cell's Own Classifier Realizes